

LEDGER · 2026-08-29 · FOOD SCIENCE / CHEMICAL ENGINEERING

High-Yield, Premium Hydrocolloids via Synergistic Ultrasound-Microwave-Assisted Extraction (UMAE) of Citrus Biowaste

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	Conventional Hot Acid Extraction (CE)
Entry application	Rapid-Set, High-Sugar Fruit Preserves (Jams, Jellies, and Marmalades)
Measured advantage	1.46-fold increase in absolute pectin yield (41.4% vs 28.3%)
Time to first revenue	12 months
Regulatory risk	Low — no material regulatory CapEx reported (\$0 estimated)

Why it failed

- Voided by: Negative control / sub-2x

Assessment

The report explicitly concedes that the primary performance metric (pectin yield) achieves only a 1.46-fold increase over conventional hot acid extraction, failing the strict 2.0x step-

change threshold. While cycle time velocity and initial CapEx (\$200,000) are highly favorable, biological limits in the feedstock prevent clearing the superiority gate.

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-08-29 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

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